MA388 Sabermetrics: Lesson 31

OPS+ and BABIP

Review

Let's look at J.D. Martinez and Aaron Judge.

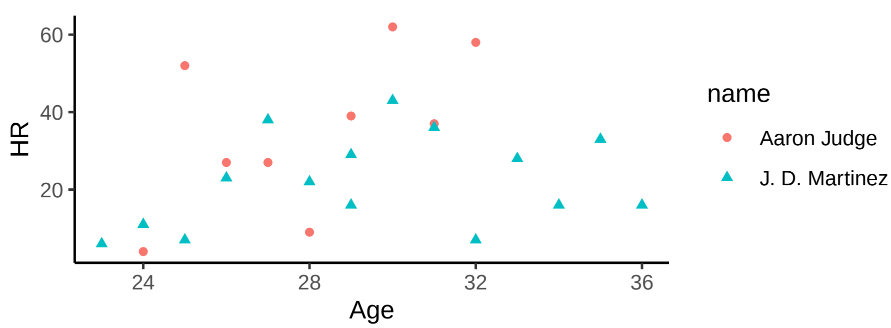
Here are their career statistics.

```
library(Lahman)
library(tidyverse)
library(knitr)

source('Chapter8_functions.R')

People |>
  filter(nameLast == "Judge", nameFirst == "Aaron") |>
  pull(playerID) -> judge_id
People |>
  filter(nameLast == "Martinez", nameFirst == "J. D.") |>
  pull(playerID) -> jd_id

player_ids <- c(judge_id, jd_id)
```

```
Batting |>
  filter(playerID %in% player_ids) |>
  group_by(playerID) |>
  summarise(HR = sum(HR),
            BB = sum(BB),
            AB = sum(AB),
            HBP = sum(HBP),
            SF = sum(SF)) |>
  mutate(PA = AB + BB + HBP + SF) |>
  left_join(People |> select(nameLast, nameFirst, playerID)) |>
  mutate(name = paste(nameFirst, nameLast, sep = " ")) |>
  select(everything(), -nameLast, -nameFirst, -HBP, -SF) -> career_totals

career_totals |>
  kable(caption = "Career Statistics (through 2022 season)")
```

Table 1: Career Statistics (through 2022 season)

playerID	HR	BB	AB	PA	name
judgeaa01	315	693	3564	4316	Aaron Judge
martijd02	331	601	6152	6854	J. D. Martinez



If Aaron Judge had the same number of plate appearances ($newPA = 5886$) as J.D. Martinez, how many career home runs would Aaron Judge have with $kicker = 1.05$?

```
newPA = 5886
career_totals |>
  filter(playerID == "judgeaa01") |>
  mutate(x = (AB/(AB + BB))*newPA - AB,
         EC = 1 + x/AB*1.05,
         AB.proj = round(x + AB),
```

```
        HR.proj = round(EC * HR)) |>
  kable()
```

playerID	HR	BB	AB	PA	name	x	EC	AB.proj	HR.proj
judgeaa01	315	693	3564	4316	Aaron Judge	1363.814	1.401797	4928	442

OPS

A common statistic reported for batters is On Base Percentage Plus Slugging (OPS).

On Base Percentage Plus Slugging = On Base Percentage (OBP) + Slugging Percentage (SLG).

On Base Percentage = $OBP = \frac{H+BB+HBP}{AB+BB+HBP+SF}$.

Slugging Percentage = $SLG = \frac{TB}{AB} = \frac{1B+2*2B+3*3B+4*HR}{AB}$

Carl Yastrzemski

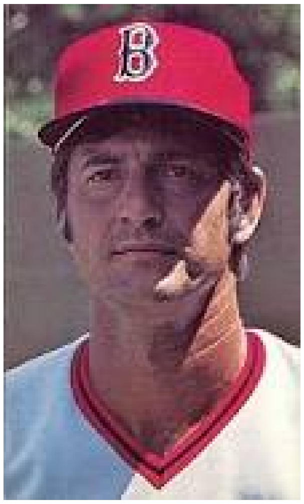
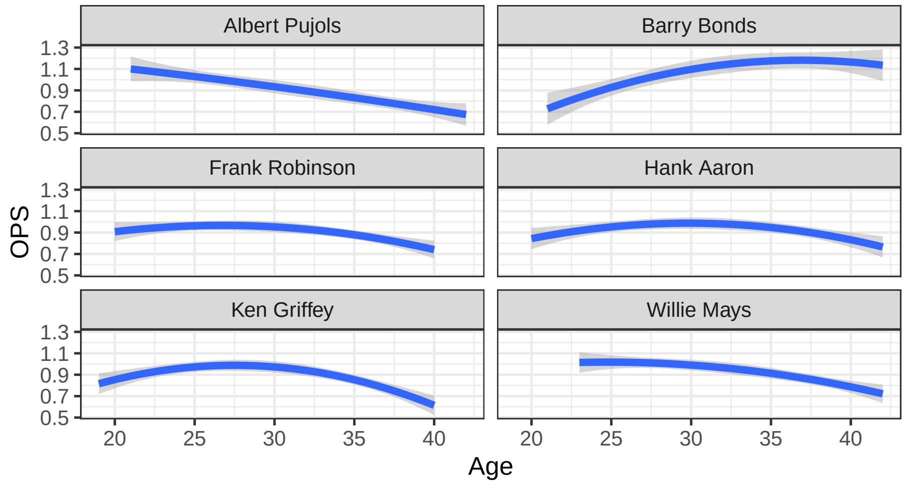


Figure 1: Yaz

Carl Yastrzemski ("Yaz") played his entire 23-year career (1961-1983) with the Boston Red Sox. Using the *similar* function of Chapter 8, here are six players with similar career statistics as Yaz.

```r
plot_trajectories("Carl Yastrzemski", n.similar = 6)
```



Albert Pujols

Albert Pujols played 22 seasons (2001 - 2022) with the St. Louis Cardinals and Los Angeles Angels.

Yaz vs. Pujols

Their career numbers are pretty similar.

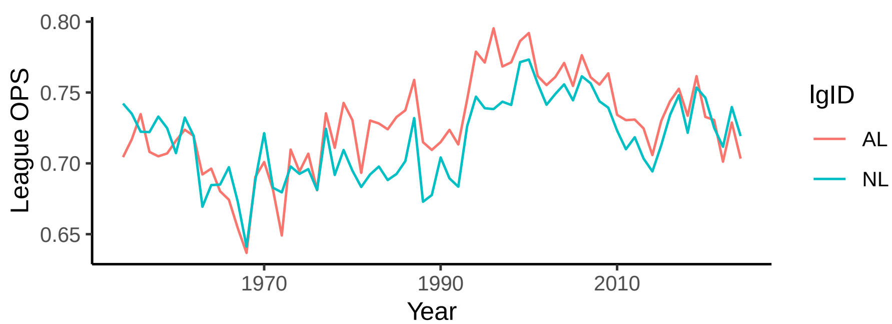
```r
player_ids <- c("yastrca01", "pujolal01")
Batting |>
```

```
  filter(playerID %in% player_ids) |>
  group_by(playerID) |>
  summarize(H = sum(H),
            AB = sum(AB),
            HR = sum(HR),
            SLG = (sum(H) - sum(X2B) - sum(X3B) - sum(HR) +
             2 * sum(X2B) + 3 * sum(X3B) + 4 * sum(HR))/sum(AB),
           OBP = (sum(H) + sum(BB) + sum(HBP))/(sum(AB) + sum(BB)+ sum(HBP) +
                                               sum(SF))) |>
  mutate(OPS = SLG + OBP,
         AVG = H / AB) |>
  left_join(People |> select(nameLast, nameFirst, playerID)) |>
  mutate(name = paste(nameFirst, nameLast, sep = " ")) |>
  select(name, everything(), -nameLast, -nameFirst) -> player_careers

player_careers |>
  kable(caption = "Career Totals", digits = 3)
```

Table 3: Career Totals

name	playerID	H	AB	HR	SLG	OBP	OPS	AVG
Albert Pujols	pujo-lal01	3384	11421	703	0.544	0.374	0.918	0.296
Carl Yastrzemski	yas-trca01	3419	11988	452	0.462	0.379	0.841	0.285

Why isn't it fair to directly compare Yaz and Pujols?



OPS+

Instead, we could use a statistic that adjusts for the overall offense in the league. OPS+ is such a statistic.

$$OPS+ = 100 \times \left(\left(\frac{OBP_{\text{player}}}{OBP_{\text{league}}} \right) + \left(\frac{SLG_{\text{player}}}{SLG_{\text{league}}} \right) - 1 \right)$$

Let's compare the OPS+ of Yaz and Pujols.

```
# Calculate OPS
Batting |>
  filter(playerID %in% player_ids) |>
  mutate(SLG = (H - X2B - X3B - HR +
             2 * X2B + 3*X3B + 4 * HR)/AB,
           OBP = (H + BB + HBP)/(AB + BB+ HBP + SF),
           OPS = SLG + OBP,
           AVG = H/AB) |>
  left_join(People |> select(nameLast, nameFirst, playerID)) |>
  mutate(name = paste(nameFirst, nameLast, sep = " ")) -> yearly_stats

# Join with league stats
yearly_stats |>
  select(name, AB, H, HR, AVG, OPS) |>
  head(5) |>
  kable(digits = 3)
```

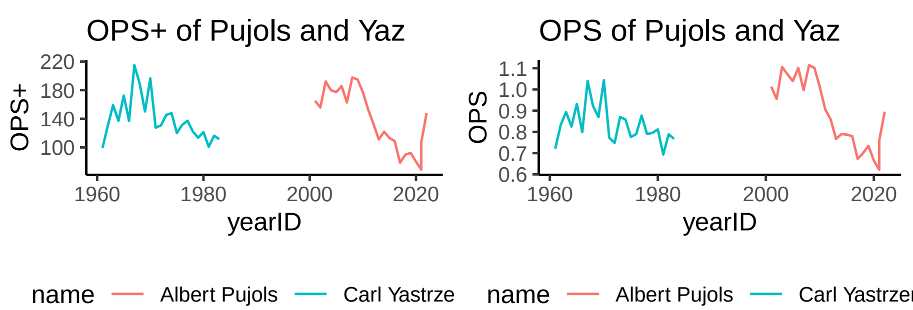
name	AB	H	HR	AVG	OPS
Albert Pujols	590	194	37	0.329	1.013
Albert Pujols	590	185	34	0.314	0.955
Albert Pujols	591	212	43	0.359	1.106
Albert Pujols	592	196	46	0.331	1.072
Albert Pujols	591	195	41	0.330	1.039

```
lg_stats |>
  head(5) |>
  kable()
```

yearID	lgID	SLG_lg	OBP_lg	OPS_lg
1954	AL	0.3733327	0.3310620	0.7043947
1954	NL	0.4070241	0.3351736	0.7421976

yearID	lgID	SLG_lg	OBP_lg	OPS_lg
1955	AL	0.3810651	0.3359929	0.7170580
1955	NL	0.4069136	0.3280744	0.7349879
1956	AL	0.3939343	0.3408003	0.7347346

```r
yearly_stats |>
  left_join(lg_stats, by = c("yearID","lgID")) -> yearly_stats

# Calculate OPS+
yearly_stats |>
  mutate(OPS_plus = 100 * ((OBP/OBP_lg + SLG/SLG_lg) - 1)) -> yearly_stats

library(gridExtra)
p1 = yearly_stats |>
  ggplot(aes(x = yearID, y = OPS_plus, color = name)) +
  geom_line() +
  theme_classic() +
  labs(y = "OPS+", title = "OPS+ of Pujols and Yaz") +
  theme(legend.position = "bottom")

p2 = yearly_stats |>
  ggplot(aes(x = yearID, y = OPS, color = name)) +
  geom_line() +
  theme_classic() +
  labs("OPS", title = "OPS of Pujols and Yaz") +
  theme(legend.position = "bottom")

grid.arrange(p1, p2, ncol = 2)
```



What are some limitations of OPS+?

Isolated Power

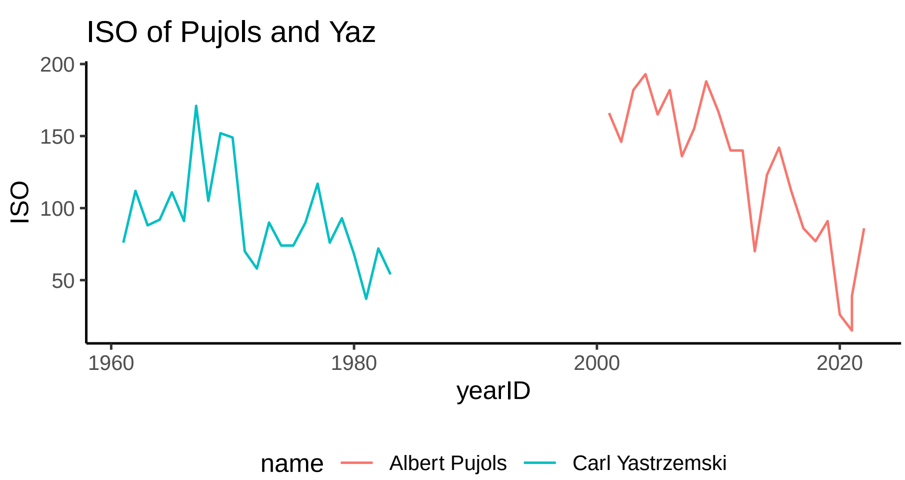
Isolated Power (ISO) is a statistic that communicates a hitter's extra bases per at-bat. Batters can provide similar offensive value with very different ISO, meaning ISO helps us understand what kind of hitter a player is more than as a standalone measure of value.

$$ISO = \frac{2B + 2 \times 3B + 3 \times HR}{AB} = SLG - AVG = \frac{\text{Extra Bases}}{\text{At-Bats}}$$

ISO is useful because two players with identical batting averages can be having very different seasons, and two players with the same slugging percentages can be having very different seasons – even if you hold walks, plate appearances, park effects, and luck constant.

Lets take a look at ISO for Albert Pujols and Yaz.

```
yearly_stats |>
  mutate(ISO = X2B + 2 * X3B + 3 * HR) |>
  ggplot(aes(x = yearID, y = ISO, color = name)) +
  geom_line() +
  theme_classic() +
  labs(title = "ISO of Pujols and Yaz")+
  theme(legend.position = "bottom")
```



BABIP (Batting Average on Balls in Play)

Why might this be an interesting metric?

What is an advantage of this metric over most others? (Hint: what players does it count for?)

BABIP Formula

$$BABIP = \frac{H - HR}{AB - SO - HR + SF}$$

Daniel Murphy Offensive Statistics (Through 2016)

The table below shows Daniel Murphy's offensive statistics for selected seasons. Note that Murphy did not play in 2010.

Year	AB	H	HR	SF	SO
2008	131	41	2	1	28
2009	508	135	12	6	69
2011	391	125	6	2	42
2012	571	166	6	4	82
2013	658	188	13	5	95
2014	596	172	9	5	86
2015	499	140	14	6	38
2016	531	184	25	8	57

Calculate Murphy's BABIP in 2009

What does that communicate to us? Who do we compare it to?